

Christopher Anaya

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SUMMARY

Full-stack software engineer building React, Go, Python, and Postgres systems for model-backed workflows. Recent work covers local-model normalization, deterministic verification, evaluation gates, review/correction flows, source inspection, and artifact-backed reports.

EXPERIENCE

Independent Open-Source Software Engineer | Mar 2026 - Present

- Built MealCheck, a hosted full-stack AI verifier with a React/Vite frontend, Go API/worker backend, Postgres run state, filesystem artifacts, Cloudflare Pages/Tunnel deployment, and private llama.cpp local-model normalization.
- Designed the AI boundary so the local model handles semantic normalization while deterministic Go code owns food resolution, guideline checks, recommendations, and final **pass** / **warn** / **block** decisions.
- Added normalized-plan review, correction, rejection, rewrite, progress, deletion, report, recommendation, and source-inspection flows backed by retained artifacts and CI-covered contracts.

Graduate Student Researcher, LASIGE / University of Lisbon | Oct 2023 - Jun 2025

- Built the engineering component of a fine-tuned biomedical QA thesis workflow and evaluated it in a public biomedical QA competition environment.
- Converted the thesis workflow into reusable evaluation tooling for seed-controlled robustness analysis and phenotype-aware report artifacts.

SELECTED PROJECTS

MealCheck

- Validated the hosted verifier with P0 normalization and P1 food/unit evaluations, live local-model smoke paths, model-comparison artifacts, Python operator summaries, and Go/TypeScript CI gates.
- Implemented source-linked review and correction before deterministic checking, then retained normalization traces, source inventory, unresolved-food summaries, recommendations, and guideline citations in completed reports.

LLM Extraction Platform

- Built an LLM backend with explicit API contracts, schema-constrained extraction, policy-aware runtime behavior, and sync/async extraction paths.
- Added durable async jobs with separate worker execution, request-trace inspection, Kubernetes deployment artifacts, and canonical evidence manifests.

TECHNICAL STRENGTHS

Languages Go, TypeScript/React, Python, SQL/Postgres, HTML, CSS

Software & Tools REST APIs, Vite, JSON Schema, llama.cpp, FastAPI, Docker, Kubernetes, Cloudflare, Redis, Prometheus/Grafana, CI/CD, Git

EDUCATION

- MSc, Data Science | Faculty of Science, University of Lisbon (2022 - 2025)
- MSc, Mathematical Research | University of Valencia (2015 - 2016)
- BA, Astronomy (Minor: Applied Mathematics) | University of Colorado Boulder (2008 - 2013)

PUBLICATIONS

- Anaya, C.; Fernandes, M.; Couto, F.M. *LLM Fine-Tuning With Biomedical Open-Source Data*. CLEF 2024 (BioASQ 12b).